

Answer Key

Question 1

Sharpe ratio = (Portfolio return – Risk-free rate) / Standard deviation

$$\text{Sharpe ratio} = (15\% - 4\%) / 11\% = 11 / 11 = 1.00$$

A Sharpe ratio of 1.00 falls in the "strong performance" range, meaning the portfolio generates one unit of excess return for every unit of total risk taken.

Question 2

Treynor ratio = (Portfolio return – Risk-free rate) / Beta

$$\text{Treynor ratio} = (12\% - 3\%) / 1.4 = 9 / 1.4 \approx 6.43$$

This means the portfolio generates about 6.43 units of excess return for each unit of systematic (market) risk taken, as measured by beta.

Question 3

$$\text{Portfolio A: Sharpe ratio} = (13\% - 2\%) / 9\% = 11 / 9 \approx 1.22$$

$$\text{Portfolio B: Sharpe ratio} = (11\% - 2\%) / 5\% = 9 / 5 = 1.80$$

Portfolio B performs better on a risk-adjusted basis. Even though Portfolio A has the higher raw return (13% vs 11%), Portfolio B achieves its return with much lower volatility (5% vs 9%). Once risk is accounted for, B delivers more excess return per unit of risk taken, making it the more efficient portfolio. This is the same lesson from the CAPM material: comparing raw returns alone can be misleading, since it ignores how much risk was taken to earn them.